	Parul University Faculty of Engineering and Technology Parul Institute of Engineering and Technology Department: AI-ML/AI-RO/AI/AI-DS/CSE/MICRO/SAP/QUICK/ORACLE/IT/AERO			
Subject Name	**PROBABILITY, STATISTICS AND NUMERICAL METHODS**	**A.Y**	**2025/2026**	
Subject Code	**303191251**	**Semester**	4^{th}	
Tutorial-4				
Sr No	**Question**	**COs**	**B.T**	**Competence**
1	Use Bisection Method to find the root of $f(x) = x^3 - 4x - 9$ in the interval (2, 3) for 5 iterations.	4	3	Apply
2	Apply Regula-Falsi Method to find a root of $f(x) = x^2 - 3x - 4$ in the interval (3, 5), correct to 4 decimal places.	4	4	Analyze
3	Use Newton-Raphson Method to find a root of $f(x) = \cos(x) - x$ starting at $x = 0.5$, perform 3 iterations.	4	4	Analyze
4	Solve $f(x) = e^x - 3x = 0$ using Newton-Raphson Method up to 3 iterations.	4	5	Evaluate
5	Use Bisection Method to find a root of $f(x) = x^3 - 2x - 5$ in the interval (2, 3) up to 5 iterations.	4	3	Apply
6	Use Regula-Falsi Method to solve $f(x) = \log_{10}(x) + x - 3 = 0$ and find the root to 4 decimal places.	4	6	Create
7	Use Newton-Raphson Method to find $\sqrt{5}$ by solving $f(x) = x^2 - 5$, starting at $x = 2$, perform 3 iterations.	4	3	Apply
8	For the equation $f(x) = x - \cos(x)$, compare the first 2 iterations of Bisection Method and Newton-Raphson Method (conceptual + numerical).	4	6	Create